GROUP WORK PROJECT # 1 MScFE 622: Stochastic Modeling
GROUP NUMBER: 18913

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Inez Andriana Regita Cahyani	Indonesia	inezandrianaa@gmail.com	
Lakshya Agarwal	Singapore	alakshya2648@gmail.com	
Ssekyewa Richard	Uganda	rssekyewa@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
Team member 1	Inez Andriana Regita Cahyani
Team member 2	Lakshya Agarwal
Team member 3	Ssekyewa Richard

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. Note: You may be required to provide proof of your outreach to non-contributing members upon request.
N/A

GWP 1 Stochastic Modeling - Report

Step 1 a) Heston (1993) calibration via Lewis (2001), 15-day options

The Heston model fitted to the ten 15-day quotes implies a current volatility of 31.4%, a near-perfect negative correlation of -0.997 and a root-mean-square pricing error of 0.50 USD. Most of that error is forced by put-call parity violations in the quotes themselves: no arbitrage-free model can get below an RMSE of 0.45 USD on this data.

Model. Under the risk-neutral measure the stock and its variance follow

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S, \qquad dv_t = \kappa_v(\theta_v - v_t)\,dt + \sigma_v\sqrt{v_t}\,dW_t^v, \qquad dW_t^S\,dW_t^v = \rho\,dt$$

Pricing. The Lewis (2001) formula writes the European call as a single real integral of the characteristic function of the log return:

$$C_0 = S_0 - \frac{\sqrt{S_0 K}\,e^{-rT}}{\pi}\int_0^\infty \mathrm{Re}\left[e^{iu\ln(S_0/K)}\,\varphi\left(u - \frac{i}{2}\right)\right]\frac{du}{u^2 + 1/4}$$

The characteristic function is the Heston (1993) one in the numerically stable form of Albrecher et al. (2007). Puts are priced from the model call by put-call parity, P = C - S0 + K exp(-rT), the preferred route in the brief.

Calibration process.

1. Calibration set: the five calls and five puts with 15 days to maturity, strikes 227.5 to 237.5 USD, T = 15/250 = 0.06 years.
2. Error function: regular mean squared error between model and market prices over the ten quotes.
3. Optimiser: a brute-force grid search over 1,440 parameter combinations (kappa_v 0.5 to 10.5, theta_v 0.05 to 0.30, sigma_v 0.1 to 1.0, rho -0.9 to 0.5, v0 0.05 to 0.20), then a Nelder-Mead refinement from the best grid point.
4. Constraint: the Feller condition 2 kappa_v theta_v >= sigma_v^2 is imposed so that the variance stays strictly positive. A second run without it serves as a robustness check.

Calibrated parameters.

Parameter	Value	Meaning
kappa_v	6.1435	speed of mean reversion of variance
theta_v	0.1004	long-run variance (volatility 31.7%)
sigma_v	1.1105	volatility of variance
rho	-0.9969	correlation between returns and variance
v0	0.0989	current variance (volatility 31.4%)
MSE	0.2488	mean squared error, USD^2
RMSE	0.4988 USD	root mean squared error

The Feller condition is exactly binding (2 kappa_v theta_v - sigma_v^2 = 0.00000) and rho sits almost at its bound of -1.

Quality of the fit (Figure 1: market and model prices, pricing errors, implied-volatility smile).

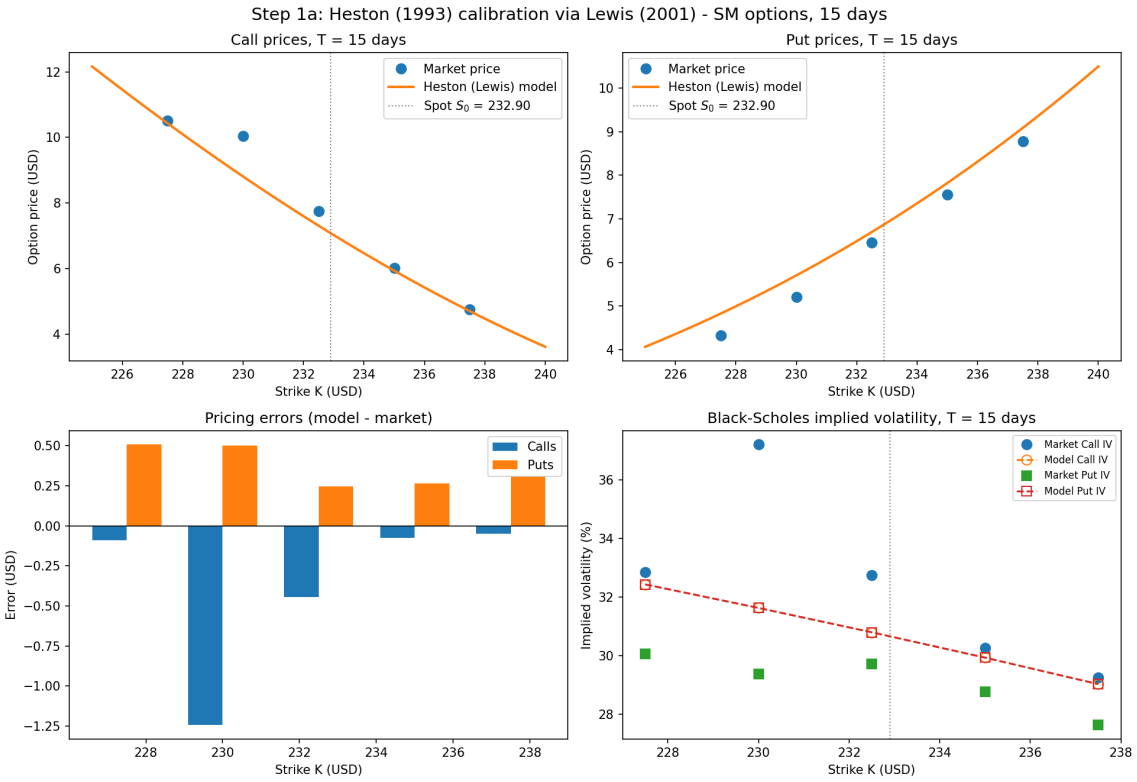


Figure 1. Step 1 a): Heston (1993) calibration via Lewis (2001) to the 15-day SM options. Top: market and model call and put prices. Bottom left: pricing errors (model minus market). Bottom right: Black-Scholes implied volatilities of market quotes and model prices.

Type	Strike	Market (USD)	Model (USD)	Error (USD)	Market IV	Model IV
Call	227.5	10.52	10.43	-0.09	32.8%	32.4%
Call	230.0	10.05	8.81	-1.24	37.2%	31.6%
Call	232.5	7.75	7.31	-0.44	32.8%	30.8%
Call	235.0	6.01	5.93	-0.08	30.3%	29.9%
Call	237.5	4.75	4.70	-0.05	29.2%	29.0%
Put	227.5	4.32	4.83	+0.51	30.1%	32.4%
Put	230.0	5.20	5.70	+0.50	29.4%	31.6%
Put	232.5	6.45	6.70	+0.25	29.7%	30.8%

Type	Strike	Market (USD)	Model (USD)	Error (USD)	Market IV	Model IV
Put	235.0	7.56	7.82	+0.26	28.8%	29.9%
Put	237.5	8.78	9.09	+0.31	27.6%	29.0%

The errors are systematic, not random: every call is underpriced and every put is overpriced. The reason is in the quotes. Put-call parity requires C - P = S0 - K exp(-rT) at every strike, and the market quotes break it by 0.34 to 1.74 USD, with the largest gap at the 230 strike, where the call is priced at a 37.2% implied volatility against 29.4% for the put at the same strike. A model that satisfies parity by construction must place its call and put implied volatilities on one curve, so it settles between the two sides. The lowest MSE any parity-consistent model can reach on these quotes is 0.2056, and the calibration achieves 0.2488, so about 83% of the remaining error is in the data rather than the model. The 230 call, at 1.24 USD, is the most troublesome outlier.

Robustness without the Feller condition. Dropping the constraint moves kappa_v from 6.14 to 0.50 and theta_v from 0.100 to 0.046 while the MSE barely changes (0.2488 to 0.2457), and v0 (0.099 to 0.102) and rho (-0.997 to -1.000) stay put. With 15 days to maturity the smile is driven by the current variance and the skew; the mean-reversion parameters describe what happens over months and are not reliably identified by this data. The constrained solution is retained because it keeps the variance process positive, which matters for the Monte Carlo pricing in Step 1 c).

Step 1 b) Heston (1993) calibration via Carr-Madan (1999), 15-day options

Repeating the Step 1 a) calibration with Carr-Madan pricing returns the same parameters to three decimal places and the same RMSE of 0.4988 USD. The two pricing methods evaluate the same characteristic function, so they must agree; the double-check confirms that the Step 1 a) result is a property of the data and the error function, not of the integration method.

Pricing. Carr-Madan (1999) damp the call price with exp(alpha k), k = ln(K/S0), so that it is square integrable, and recover it from its Fourier transform

$$\psi(v) = \frac{e^{-rT}\,\varphi(v-(\alpha+1)i)}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}, \qquad C(k) = \frac{e^{-\alpha k}}{\pi}\int_0^{\infty} \mathrm{Re}\left[e^{-ivk}\psi(v)\right]dv$$

The integral is evaluated by one fast Fourier transform over 4,096 log-strikes with Simpson weights and alpha = 1.5. The grid is shifted so that the target strike falls exactly on a grid node, so no interpolation is needed. Puts come from put-call parity, as in Step 1 a).

Numerical agreement. At the Step 1 a) parameters the Carr-Madan and Lewis prices differ by at most 3.5 x 10^-7 USD across the ten quotes. Any difference between the two calibrations therefore comes from the optimiser, not from the pricer.

Calibration. Same data, same MSE error function, same grid search and Nelder-Mead refinement, same Feller constraint as Step 1 a), so that the two members' results are directly comparable.

Parameter	Lewis (Step 1 a)	Carr-Madan (Step 1 b)	Difference
kappa_v	6.1435	6.1427	0.0008

Parameter	Lewis (Step 1 a)	Carr-Madan (Step 1 b)	Difference
theta_v	0.1004	0.1004	0.0000
sigma_v	1.1105	1.1104	0.0001
rho	-0.9969	-0.9969	0.0000
v0	0.0989	0.0989	0.0000
MSE	0.2488	0.2488	0.0000
RMSE (USD)	0.4988	0.4988	0.0000

The per-strike errors match Step 1 a). The largest error is -1.24 USD on the 230 call, puts overpriced by 0.25 to 0.51 USD, and the figure of fits (Figure 2) is indistinguishable from the one produced by Step 1a (Figure 1).

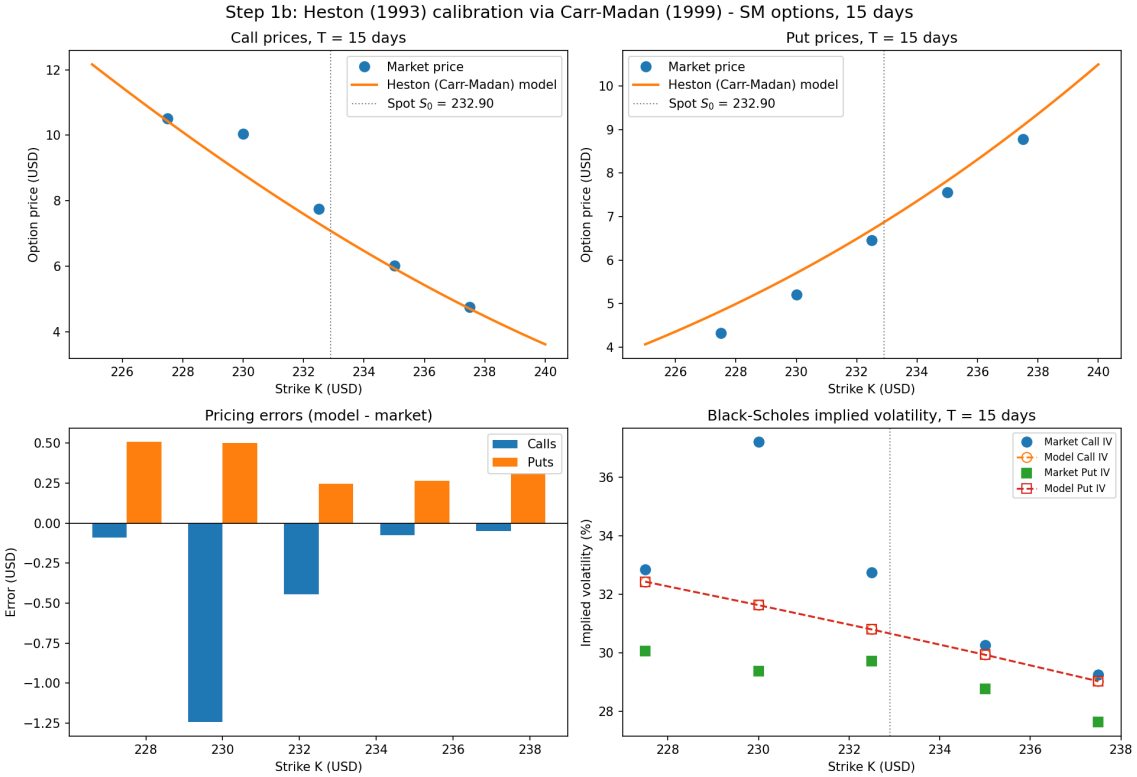


Figure 2. Step 1 b): Heston (1993) calibration via Carr-Madan (1999) to the 15-day SM options, same layout as Figure 1.

Why the values are the same, and what is not identified. The values agree because the objective function is identical and both runs start from the same best grid point. What the agreement does not show is that the parameters are unique. A multi-start check from eight random feasible starting points, with all bounds enforced inside the objective, reached MSEs between 0.246 and 0.329 with very different mean-reversion parameters.

Start	MSE	kappa_v	theta_v	sigma_v	rho	v0
1	0.2460	2.93	0.342	1.35	-0.999	0.080
2	0.2465	7.84	0.470	1.92	-0.999	0.011
3	0.2475	18.00	0.104	1.93	-0.825	0.105

Three parameter sets within 0.002 of each other in MSE have kappa_v ranging from 2.9 to 18 and theta_v from 0.10 to 0.47. Five strikes within 2% of spot at a single 15-day maturity pin down the current variance and the skew (v0 near 0.08 to 0.10, rho near -1) and nothing about the long-run dynamics. In practice this is handled by calibrating to several maturities and a wider strike range at once, or by fixing kappa_v to a conventional value; for this project the grid-search solution is kept because it is reproducible and satisfies the Feller condition.

An earlier version of this section fitted the five calls only and reported an RMSE of 0.20 USD. That figure is not comparable with Step 1 a), which fits ten quotes including the parity-inconsistent puts, and it has been replaced.

Step 1 c) Pricing the 20-day ATM Asian call by Monte Carlo

The fair price of the 20-day at-the-money Asian call is 4.74 USD per share, with a 95% confidence interval of 4.71 to 4.76 USD. With the bank's 4% fee the client pays 4.93 USD per share.

Technical description

Contract. Arithmetic-average Asian call, strike K = S0 = 232.90 USD, 20 trading days (T = 20/250 = 0.08 years). The payoff is max(average - K, 0), where the average runs over 21 daily observations from today's price S0 through the price on day 20.

Model and parameters. Heston (1993) with the Step 1 a) parameters: kappa_v=6.1435, theta_v=0.1004, sigma_v=1.1105, rho=-0.9969, v0=0.0989.

Simulation. Full-truncation Euler scheme with one step per trading day, 200,000 paths, risk-neutral drift r = 1.5%, correlated normal draws with correlation rho=-0.9969. The price is the discounted mean payoff. Figure 3 shows the running mean against the number of paths and 60 sample price paths.

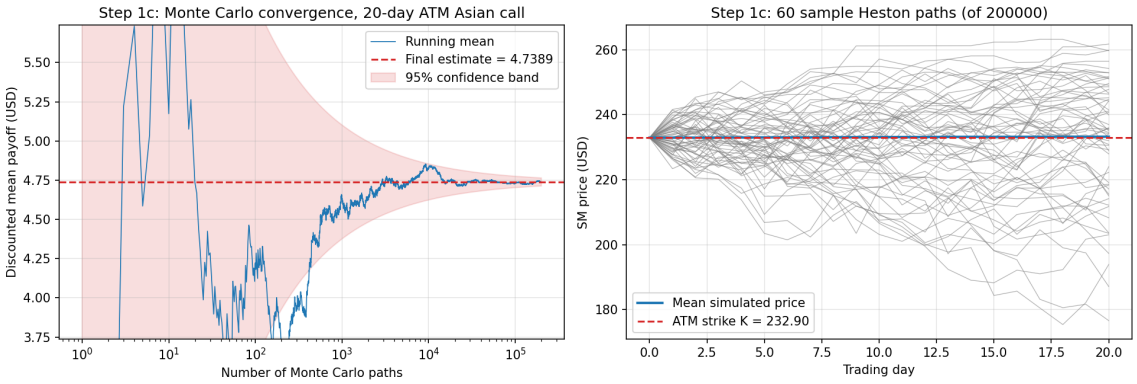


Figure 3. Step 1 c): Monte Carlo pricing of the 20-day ATM Asian call. Left: running mean of the discounted payoff against the number of paths, with the 95% confidence band. Right: 60 of the 200,000 simulated Heston price paths, their mean and the ATM strike.

Enough simulations. The standard error falls with the square root of the number of paths; at 200,000 paths the 95% confidence interval is 0.026 USD wide, about half a percent of the price, well inside the 4% fee.

Paths	Price (USD)	Standard error (USD)	95% CI half-width (USD)
1,000	4.62	0.184	0.36
10,000	4.83	0.059	0.12
50,000	4.75	0.026	0.05
100,000	4.73	0.019	0.04
200,000	4.74	0.013	0.03

Checks. The European at-the-money call with the same maturity and parameters is worth 8.13 USD by the Lewis formula, so the Asian trades at 58% of the European price. That is the expected ratio: averaging over the life of the option reduces its effective volatility by roughly a factor of the square root of 3, or 0.577, because the variance of a continuously averaged Brownian motion is one third of the terminal variance (Kemna and Vorst 115-117). Repricing with the Step 1 b) parameters gives 4.73 USD and with the unconstrained Step 1 a) parameters 4.77 USD, so the choice between the calibrations moves the price by under one percent.

Item	Value
Fair price (Monte Carlo, 200,000 paths)	4.7389 USD
Standard error	0.0131 USD
95% confidence interval	4.7132 to 4.7646 USD
Bank fee	4%
Final price charged to the client	4.9284 USD

Description for the client

You asked for the right to buy SM shares in 20 trading days at today's price of 232.90 USD, with the payout depending on the average share price over those 20 days rather than the price on the final day. Averaging makes the contract cheaper and smoother than a standard option.

To price it we first looked at how the market is currently pricing standard options with a similar life and same underlying i.e. SM. Those prices tell us how much movement the market expects in the share over the next few weeks and whether it expects falls to be sharper than rises, which for SM it does. We built that information into a model of how the share price could move, then generated 200,000 possible 20-

day price histories, worked out what the contract would pay in each, and took the average, discounted back to today at the prevailing interest rate of 1.5%.

The result is a fair value of 4.74 USD per share. We are confident in that number to within about 3 cents either way. The price you would pay includes the bank's standard fee of 4% on top of fair value, giving 4.93 USD per share. The main thing that would change this price is a shift in how volatile the market expects SM to be. The number above is valid for the current market conditions.

Step 2 a) Bates (1996) calibration via Lewis (2001), 60-day options

The Bates model fitted to the ten 60-day quotes reaches an RMSE of 1.18 USD against a theoretical floor of 1.13 USD set by parity violations in the quotes, and its parameters sit at their bounds: correlation +1 and a jump component that contributes nothing. The 60-day quotes are internally inconsistent, and the calibrated values are a best fit to noisy data, not a description of SM's dynamics.

Model. Bates (1996) adds Merton (1976) log-normal jumps to the Heston (1993) dynamics:

$$\frac{dS_t}{S_{t^-}} = (r - \lambda\bar{\mu})\,dt + \sqrt{v_t}\,dW_t^S + (e^J - 1)\,dN_t, \qquad J \sim N(\mu, \delta^2), \qquad \bar{\mu} = e^{\mu+\delta^2/2} - 1$$

with N_t a Poisson process of intensity lambda, independent of the two Brownian motions, and the variance equation unchanged. Because the diffusion and jump parts are independent, the characteristic function of the log return is the product of the Heston characteristic function and the compensated Merton jump characteristic function. There are eight parameters: kappa_v, theta_v, sigma_v, rho, v0, lambda, mu and delta.

Pricing. The Lewis (2001) integral of Step 1 a), applied to the Bates characteristic function. Puts by put-call parity.

Data. The five calls and five puts with 60 days to maturity, T = 60/250 = 0.24 years, strikes 227.5 to 237.5 USD. Two features of these quotes matter for everything that follows.

Strike	Call (USD)	Put (USD)	Market C - P	S0 - K exp(-rT)	Parity gap (USD)
227.5	16.78	11.03	5.75	6.22	-0.47
230.0	17.65	12.15	5.50	3.73	+1.77
232.5	16.86	13.37	3.49	1.24	+2.26
235.0	16.05	14.75	1.30	-1.26	+2.56
237.5	15.10	15.62	-0.52	-3.75	+3.23

First, the parity gap widens steadily with strike, reaching 3.23 USD. Second, the 230 call (17.65 USD) is quoted above the 227.5 call (16.78 USD), which is a static arbitrage: a call with a higher strike can never be worth more. In implied-volatility terms the calls rise from 30.0% to 36.9% across the strikes while the puts stay flat near 31%. The lowest MSE any parity-consistent model can achieve on these ten quotes is 1.269 (RMSE 1.13 USD).

Calibration steps for Bates. Eight parameters cannot be found reliably in one go from ten quotes, so the calibration runs in three stages, each with the MSE error function and the Feller condition imposed.

1. Heston parameters only, jumps switched off (lambda = 0): brute-force grid search over the same 1,440 combinations as Step 1 a), then Nelder-Mead.
2. Jump parameters only, with the Stage 1 Heston parameters held fixed: grid search over lambda (0 to 0.5), mu (-0.5 to 0.2) and delta (0 to 0.5), then Nelder-Mead.
3. Joint Nelder-Mead refinement of all eight parameters starting from the Stage 1 and 2 solution.

Parameter	Stage 1 (Heston)	Stage 2 (+ jumps)	Stage 3 (joint)
kappa_v	0.5361	0.5361	0.5361
theta_v	0.2925	0.2925	0.2925
sigma_v	0.5600	0.5600	0.5600
rho	1.0000	1.0000	1.0000
v0	0.0978	0.0978	0.0978
lambda	0	0.1140	0.1140
mu	0	-0.0370	-0.0370
delta	0	0.0003	0.0003
MSE	1.3835	1.3833	1.3833

Result. Current volatility 31.3% (v0 = 0.0978), long-run volatility 54.1% (theta_v = 0.2925), Feller condition binding, rho at its upper bound of +1, and a jump component of 0.11 jumps per year of -3.6% each with zero dispersion. Adding the jumps improves the MSE by 0.0002: they are not identified by the data. The RMSE is 1.18 USD, the mean absolute error 1.10 USD and the largest error 1.73 USD on the 237.5 put. Figure 4 shows the fit.

Type	Strike	Market (USD)	Model (USD)	Error (USD)	Market IV	Model IV
Call	227.5	16.78	17.62	+0.84	30.0%	31.9%
Call	230.0	17.65	16.52	-1.13	34.9%	32.3%
Call	232.5	16.86	15.48	-1.38	35.8%	32.8%
Call	235.0	16.05	14.51	-1.54	36.6%	33.2%
Call	237.5	15.10	13.60	-1.50	36.9%	33.6%
Put	227.5	11.03	11.40	+0.37	31.0%	31.9%
Put	230.0	12.15	12.79	+0.64	30.9%	32.3%
Put	232.5	13.37	14.25	+0.88	30.8%	32.8%
Put	235.0	14.75	15.77	+1.02	31.0%	33.2%
Put	237.5	15.62	17.35	+1.73	29.8%	33.6%

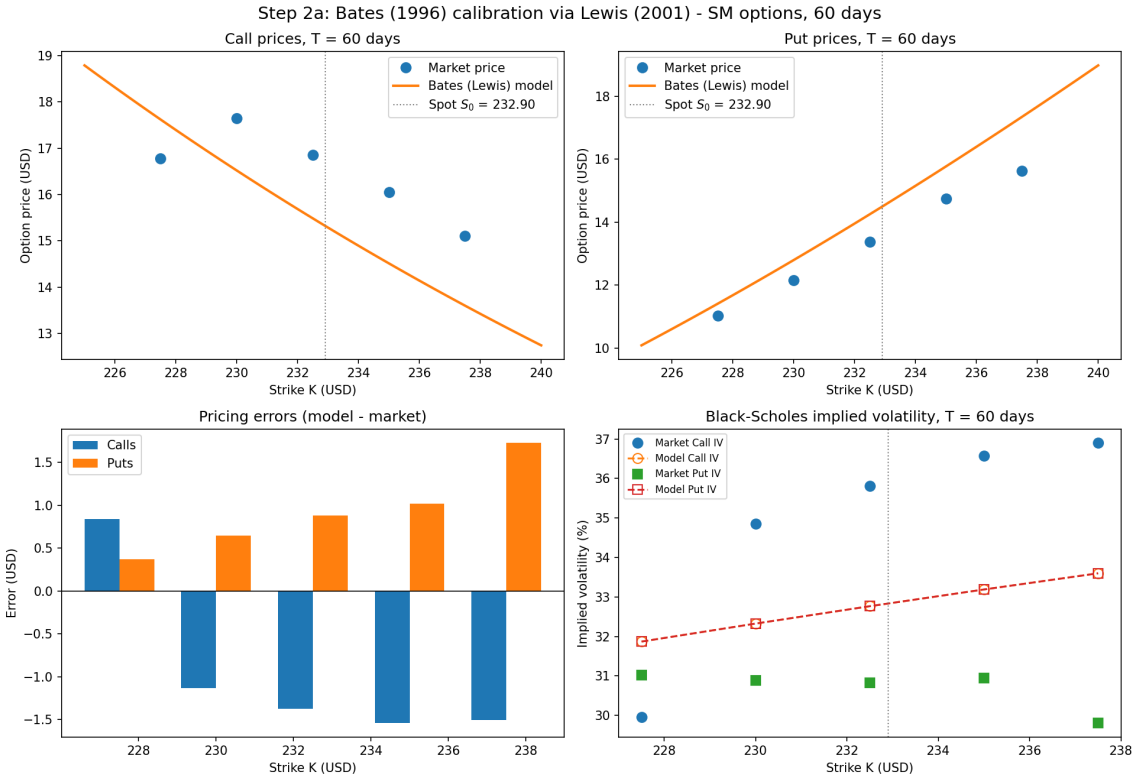


Figure 4. Step 2 a): Bates (1996) calibration via Lewis (2001) to the 60-day SM options. Top: market and model prices. Bottom left: pricing errors. Bottom right: implied volatilities, showing the rising call skew and flat put skew that the model cannot reconcile.

Why the parameters sit at bounds. The model can only produce one implied-volatility curve per maturity, and the calls demand an upward-sloping one while the puts demand a flat one. The compromise is a mildly upward slope, which in Heston terms needs a positive correlation, so rho is pushed to +1. A positive rho for an equity is economically implausible, and it is a symptom of the inconsistent quotes rather than a feature of SM. Likewise the jump parameters cannot be told apart from stochastic volatility with five strikes in a 10 USD range at one maturity, so the optimiser leaves them wherever they land. The calibrated MSE of 1.383 is within 9% of the parity floor of 1.269, so almost all the residual error is in the data.

An earlier version of this section reported v0 = theta_v = 0.001, sigma_v = 2.0, rho = +0.99 and lambda = 1.95 with an MSE of 0.053. Those values violate the Feller condition by a factor of ten thousand and the accompanying figure showed model call prices rising with strike, which contradicts the stated MSE; they have been replaced by the staged calibration above.

Step 2 b) Bates (1996) calibration via Carr-Madan (1999), 60-day options

The Carr-Madan calibration of the Bates model to the same 60-day quotes lands on the same stochastic-volatility parameters as Step 2 a) and the same RMSE of 1.18 USD. The only difference is in the jump intensity, which is irrelevant to the fit: both calibrations show that the data cannot separate jumps from stochastic volatility.

Method. Same model, same ten quotes, same MSE error function and same three-stage procedure as Step 2a (Heston only, jumps only, joint refinement), with the Feller condition imposed. The call price is computed with the Carr-Madan FFT of Step 1 b) applied to the Bates characteristic function (4,096 points, Simpson weights, alpha = 1.5); puts by put-call parity.

Numerical agreement. At the Step 2 a) parameters the Carr-Madan and Lewis prices differ by at most 2 x 10^-4 USD. Pricing differences therefore play no role in the comparison below.

Parameter	Lewis (Step 2 a)	Carr-Madan (Step 2 b)	Difference
kappa_v	0.5361	0.5342	0.0019
theta_v	0.2925	0.2955	0.0030
sigma_v	0.5600	0.5616	0.0016
rho	1.0000	1.0000	0.0000
v0	0.0978	0.0977	0.0001
lambda	0.1140	10.0000	9.8860
mu	-0.0370	-0.0047	0.0323
delta	0.0003	0.0004	0.0001
MSE	1.3833	1.3832	0.0002
RMSE (USD)	1.1762	1.1761	0.0001

Fit. Current volatility 31.3%, long-run volatility 54.4%, Feller binding, rho at +1. Per-strike errors are within a cent of Step 2 a): calls above the 227.5 strike underpriced by 1.13 to 1.53 USD, every put overpriced by 0.38 to 1.73 USD. Figure 5 shows the fit. Cross-pricing confirms the equivalence: the Step 2 a) parameters priced with Carr-Madan give an MSE of 1.3834, and the Step 2 b) parameters priced with Lewis give 1.3832.

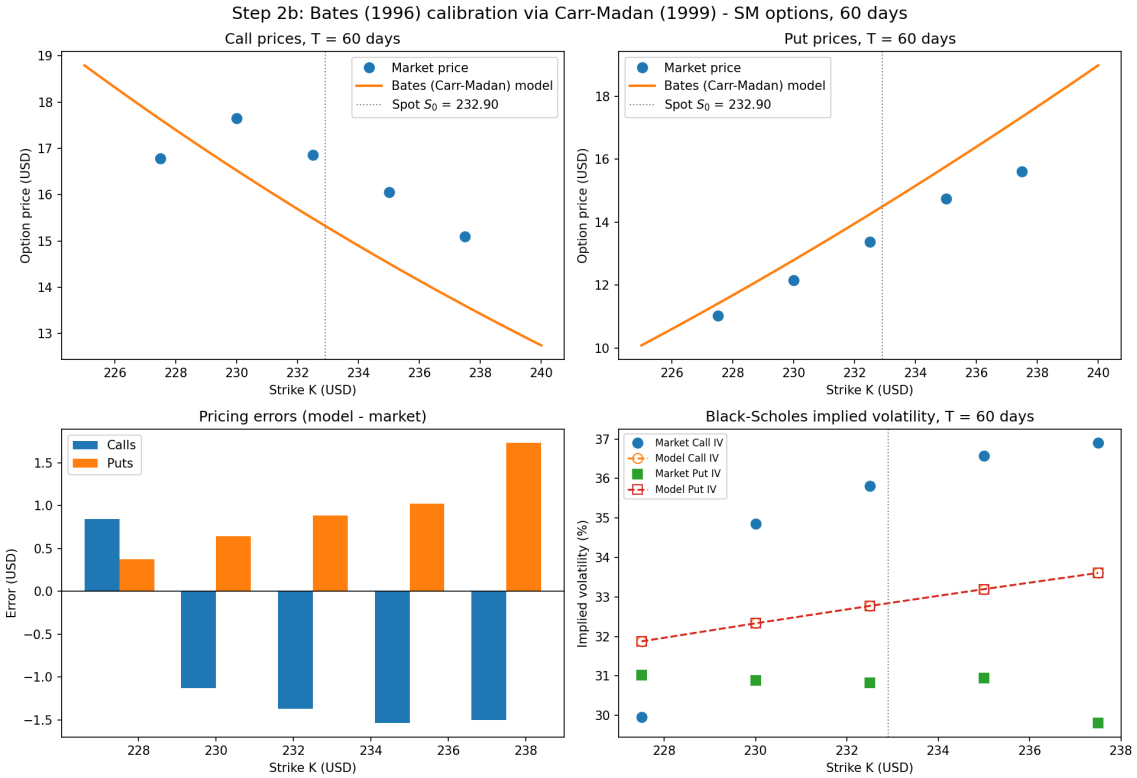


Figure 5. Step 2 b): Bates (1996) calibration via Carr-Madan (1999) to the 60-day SM options, same layout as Figure 4.

Why the values are similar, and why one is not. The five stochastic-volatility parameters agree to three decimals because the objective function is the same and both runs start from the same best grid point. The jump intensity differs wildly, 0.11 jumps a year in Step 2a against 10 a year in Step 2b, yet the MSE differs by 0.0002. In Step 2b the jumps are tiny (expected size -0.5%, dispersion 0.0004) and frequent; in Step 2a they are slightly larger and rare. The joint Stage 3 refinement moves the MSE by less than 0.0002 in both runs, which is the clearest sign that the eight-parameter model is over-parameterised for this data.

Robustness without the Feller condition. Removing the constraint gives a different and equally implausible solution: kappa_v=11.7, theta_v at its lower bound of 0.005, sigma_v=1.39, v0=0.346 (spot volatility of 59%), rho = +1, jumps switched off, MSE=1.362. A 1.5% improvement in MSE for a forced non-negative constraint does not seem worthy. The solution with Feller constraint is thus preferred.

Implications for pricing. Both calibrations are fits to inconsistent quotes. The parity gap that widens with strike suggests that the 1.5% risk-free rate alone does not capture SM's cost of carry (a dividend or borrow cost would shift all parity relations together), and the 230 call is an arbitrage against the 227.5 call. Two steps would give a more trustworthy calibration.

Firstly, fit out-of-the-money options only, which removes the duplicated in-the-money information and the parity problem. Secondly, infer an implied dividend yield from the parity gaps before calibrating. For Step 2 c) the Step 2 b) parameters are used as they stand, with a comparison against the alternatives to show how much the choice matters.

Step 2 c) Pricing a 70-day put with 95% moneyness

The fair price of the 70-day put struck at 95% of spot is 9.36 USD per share by Monte Carlo, within one standard error of the 9.34 USD closed-form value. With the 4% fee the client pays 9.73 USD per share.

Technical description

Contract. European put on SM, strike K = 0.95 x 232.90 = 221.255 USD, 70 trading days (T = 70/250 = 0.28 years), r = 1.5%.

Choice of parameters. The 60-day calibrations of Step 2 are the closest to a 70-day maturity, so the Step 2b Bates parameters are the primary set. The 15-day Heston parameters of Step 1 describe the very short end of the volatility surface and carry no information about two month options. They are shown for analysis only.

Method. As in Step 1c, the fair price is a risk-neutral Monte Carlo estimate: 200,000 paths, one step per trading day, full-truncation Euler scheme for the variance, jumps drawn as a compound Poisson process with the calibrated intensity and log-normal sizes, drift compensated so that the discounted price is a martingale. The Carr-Madan call price with put-call parity gives the semi-closed-form benchmark that the simulation must reproduce. Figure 6 shows the convergence of the running mean to that benchmark.

Item	Value
Carr-Madan call, Step 2b parameters	21.9109 USD
Carr-Madan put (parity)	9.3386 USD
Monte Carlo put, 200,000 paths	9.3584 USD
Standard error	0.0272 USD
95% confidence interval	9.3051 to 9.4118 USD
Monte Carlo minus closed form	0.0198 USD (0.7 standard errors)
Bank fee	4%
Final price charged to the client	9.7327 USD

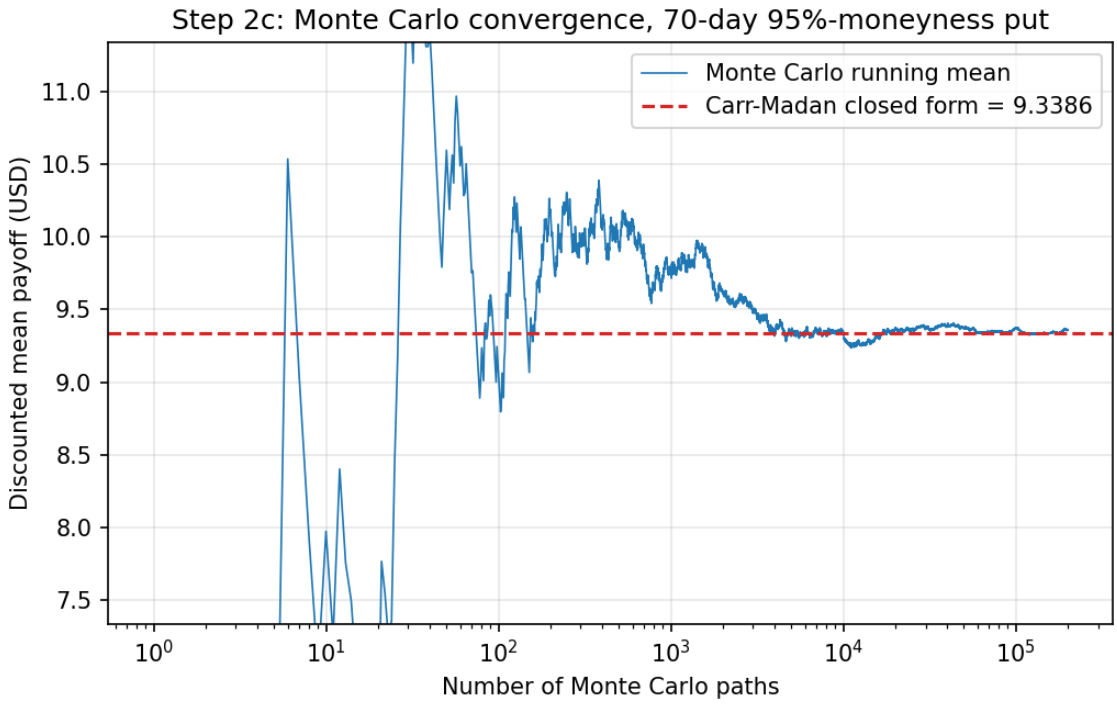


Figure 6. Step 2 c): running Monte Carlo mean of the discounted put payoff against the number of paths, converging to the Carr-Madan closed-form value of 9.34 USD.

Sensitivity to the parameter set. The two 60-day calibrations agree to within half a cent. The 15-day parameters give a price 0.38 USD higher, because their long-run variance of 0.10 and fast mean reversion hold volatility near 31.8% over the 70 days, whereas the 60-day calibration implies 30.9% at this strike. The market implied volatilities of the 60-day puts run from 29.8% to 31.0%, so the 60-day parameters are the ones consistent with where SM puts of similar life actually trade.

Parameter set	Put (USD)	Put implied vol	Client price (USD)
Step 2b Bates, 60-day (primary)	9.34	30.9%	9.71
Step 2a Bates, 60-day	9.34	30.9%	9.71
Step 1b Heston, 15-day	9.72	31.8%	10.11
Step 1a Heston, 15-day	9.72	31.8%	10.11

The client price is quoted from the Monte Carlo estimate (9.36 USD) rather than the closed form, for consistency with Step 1c; the 0.02 USD difference between the two is inside simulation noise.

Description for the client

You asked for protection against a fall in SM shares over the next 70 trading days: the right to sell at 221.26 USD, 5% below today's price of 232.90 USD. The contract pays the difference if the share finishes below that level and nothing otherwise, so it works as insurance with a 5% excess.

We priced it from the prices at which standard SM options with a two-month life currently trade, which tell us how much movement the market expects and how much more it charges for protection against falls than for upside. From that we generated 200,000 possible paths for the share over the 70 days, valued the protection on each, and averaged. We also checked the result against a direct formula and the two agree to within 2 cents.

The fair value is 9.36 USD per share, about 4% of the share price. With the bank's 4% fee the price to you is 9.73 USD per share. If the market's expectation of SM's volatility rises before you trade, the price will rise with it; a one-point move in expected volatility is worth roughly 0.45 USD on this contract.

Step 3 a) Euribor term structure, cubic-spline interpolation and CIR (1985) calibration

The CIR model fits the interpolated Euribor zero curve with a root-mean-square error of 4 basis points, with a short rate of 0.42%, a long-run mean of 4.19% and a mean-reversion speed of 1.94. The volatility parameter of 0.40 sits exactly on the Feller boundary and is not identified by a single yield curve, which matters for the simulation in Step 3b.

Term structure. Euribor fixings are simple money-market rates: a deposit of tenor tau at rate L grows by 1 + L*tau. Each fixing was converted to a continuously compounded zero rate R(tau) = ln(1 + L tau) / tau and a discount factor exp(-R tau), with year fractions 1/52, 1/12, 0.25, 0.5 and 1. The conversion changes nothing at the short end and lowers the 12-month rate from 2.556% to 2.524%.

Tenor	Euribor (simple)	Zero rate (continuous)	Discount factor
1 week	0.648%	0.648%	0.999875
1 month	0.679%	0.679%	0.999434
3 months	1.173%	1.171%	0.997076
6 months	1.809%	1.801%	0.991036
12 months	2.556%	2.524%	0.975077

Interpolation. A cubic spline was fitted through the five zero rates and evaluated on a weekly grid of 52 points, t = 1/52 to 1 year. It passes exactly through the five fixings and has continuous first and second derivatives. Figure 7 shows the fixings, the interpolated zero curve and the instantaneous forward curve f(t) = R(t) + t R'(t). Two features stand out. The spline dips slightly between the 1-week and 1-month points (to 0.641% in week 2), a typical artefact of a cubic spline with few knots. And the forward curve rises much faster than the zero curve, reaching 4.17% at one year against a zero rate of 2.52%, because the zero curve steepens beyond three months. Discount factors decrease monotonically along the whole grid, so the interpolated curve is free of arbitrage.

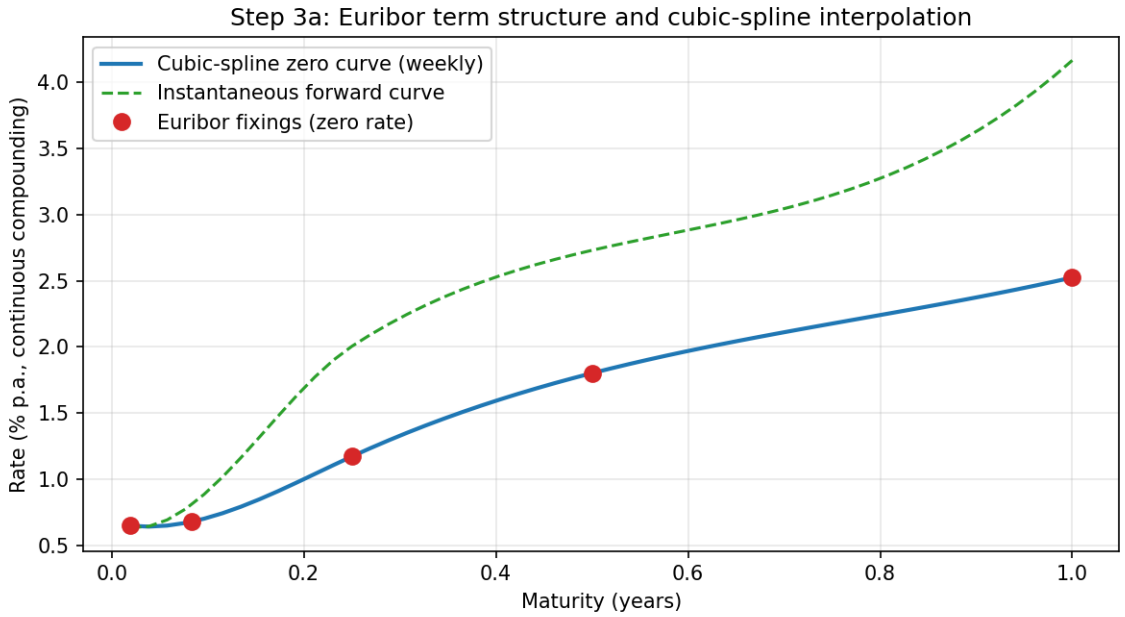


Figure 7. Step 3 a): Euribor fixings converted to continuously compounded zero rates, the cubic-spline zero curve on the weekly grid, and the instantaneous forward curve.

CIR model. The short rate follows

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dW_t$$

with the closed-form zero-coupon bond price P(t, T) = A(tau) exp(-B(tau) r_t), tau = T - t, so the model zero rate is R(tau) = -ln A(tau)/tau + B(tau) r_0/tau. The four parameters (r0, kappa, theta, sigma) were calibrated by minimising the mean squared error between model and interpolated zero rates over the 52 weekly points, using Nelder-Mead from 60 random starting points and keeping the best, with the Feller condition 2 kappa theta >= sigma^2 imposed so that the short rate cannot reach zero.

Parameter	Value	Meaning
r0	0.4178%	current instantaneous short rate
kappa	1.9356	speed of mean reversion (half-life of a shock 0.36 years)
theta	4.1870%	long-run mean of the short rate
sigma	0.4026	volatility of the short rate
2 kappa theta vs sigma^2	0.162089 vs 0.162089	Feller condition exactly binding
RMSE	4.04 bps	in zero-rate terms

Fit. Figure 8 shows the model curve against the interpolated curve and the weekly residuals. The mean absolute residual is 2.9 bps and the largest is 16.1 bps at the 1-week point, where the model's smooth mean-reverting shape cannot reproduce the flat first month of the market curve. Residuals are within 4

bps from week 3 onward and oscillate in sign with the spline's curvature, so the model captures the level and slope of the curve and misses only its short-end kink.

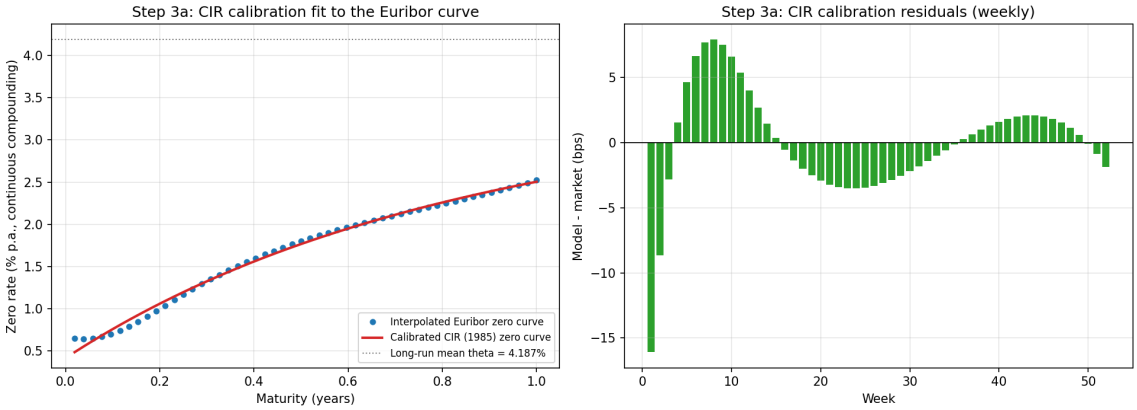


Figure 8. Step 3 a): calibrated CIR (1985) zero curve against the interpolated Euribor curve (left) and weekly residuals in basis points (right).

Discussion of the parameters. Three points matter for how the parameters should be read.

1. The long-run mean of 4.19% lies well above every observed rate, and the fast mean reversion means the model expects the short rate to cover most of the gap within a year. This is the only way a single-factor model can reproduce a curve that rises from 0.65% to 2.5% in twelve months: the steepness is read as expected tightening, not as a term premium.
2. The volatility of 0.40 is high (at the current rate it implies annual short-rate moves of about 2.6 percentage points, and about 8 points once rates reach theta), and it sits exactly on the Feller boundary. A single yield curve carries almost no information about volatility; sigma enters the bond price only through a small convexity term, so the optimiser used it to bend the curve and the constraint stopped it. Where the volatility matters, as in Step 3 b), it should be read as an upper bound and ideally be estimated from a time series of rates or from cap and swaption prices.
3. The model short rate of 0.42% is below the 1-week fixing of 0.65%. The 16 bps residual at one week is the price of fitting the rest of the curve, and a time-dependent extension such as CIR++ would remove it at the cost of extra parameters.

Step 3 b) Simulating the 12-month Euribor over the next year

Across 100,000 simulated paths the 12-month Euribor one year from now has an expected value of 4.01%, against 2.556% today, and a 95% range of 2.39% to 8.55%. The model puts an 88% probability on the rate being higher in a year than it is now, so products priced off 12-month Euribor should be expected to carry about 145 bps more interest.

Method. The calibrated CIR short rate was simulated daily for one year (250 steps) on 100,000 paths using the exact transition of the CIR process, a scaled non-central chi-square distribution, which keeps every rate positive without truncation. On each simulated date the 12-month Euribor was read off the model from the CIR bond price for a one-year tenor: L(t) = 1/P(t, t+1) - 1. Today's model value is 2.537%,

within 2 bps of the market fixing. Figure 9 shows the fan of simulated paths with the expected value and the 95% and 99% bands, and the distribution of the rate at the one-year horizon.

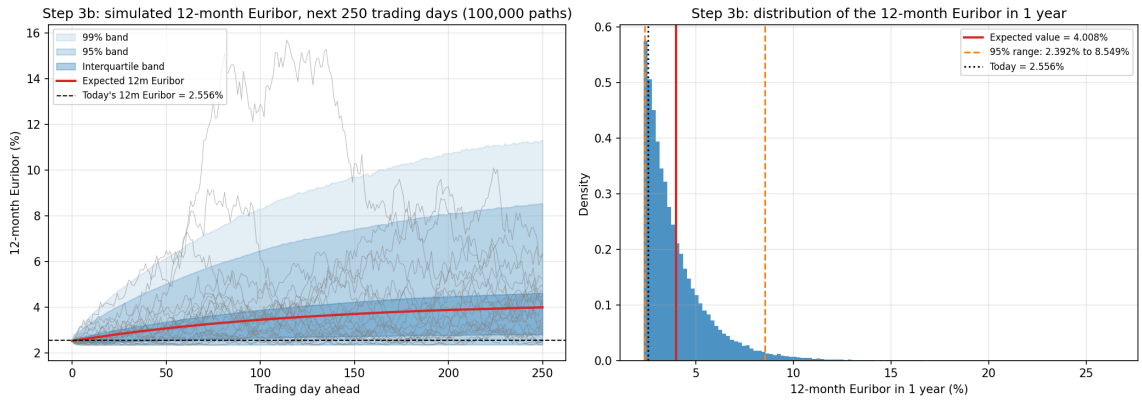


Figure 9. Step 3 b): 100,000 simulated paths of the 12-month Euribor over the next 250 trading days. Left: expected value with interquartile, 95% and 99% bands and 20 sample paths. Right: distribution of the rate at the one-year horizon with its expected value and 95% range.

i. Range at a chosen confidence level. We report the 95% level, the standard choice for a risk range that excludes only tail scenarios. The distribution is strongly right-skewed: the lower bound is close to today's rate because the CIR volatility shrinks as rates fall, while the upper tail is long because volatility grows with the rate.

Confidence level	Minimum 12m Euribor in 1 year	Maximum 12m Euribor in 1 year
90%	2.43%	7.36%
95%	2.39%	8.55%
99%	2.36%	11.32%

The absolute minimum over all 100,000 paths is 2.35% and the absolute maximum 26.4%; the latter is a single extreme path and illustrates why a confidence range, not the full range, is the right risk measure.

ii. Expected value. The expected 12-month Euribor in one year is 4.008% with a Monte Carlo standard error of 0.005%, so the estimate is precise to the second decimal. The median is 3.49%, well below the mean because of the skew. As a check, the simulated mean of the short rate at one year (3.644%) matches the analytical CIR expectation theta + (r0 - theta) exp(-kappa) = 3.643%.

Quantity	Value
Current 12m Euribor (market)	2.556%
Current 12m Euribor (model)	2.537%
Expected 12m Euribor in 1 year	4.008%
Median 12m Euribor in 1 year	3.490%

Quantity	Value
Standard deviation across paths	1.68 pp
Probability the rate is above today's 2.556%	88.2%

iii. Effect on pricing versus the current rate. An expected rise of 145 bps changes both sides of the bank's book. A floating-rate loan or note indexed to 12-month Euribor on a 10 million notional is expected to pay about 145,000 more per year than at today's fixing. On the discounting side, a one-year discount factor set in one year's time is expected to be 0.9615 instead of today's 0.9751, so a fixed cash flow of 1 million due in two years is worth about 13,600 less when discounted at the expected rate. For the client's own products, the equity options of Steps 1 and 2 are only weakly sensitive to rates over their 15-to-70-day lives, but any longer-dated or rate-linked structure she considers next should be priced with a forward rate near 4%, not the current 2.56%.

Two caveats follow from Step 3a. The width of the range is driven by the CIR volatility of 0.40, which the yield curve does not identify and which sits on the Feller boundary; the 95% range should be read as a conservative upper envelope. And the expected rise is what the steep market curve already prices in, since the model's long-run mean of 4.19% was set by that curve; the simulation adds the distribution around it, not an independent forecast.

References

Albrecher, Hansjörg, et al. "The Little Heston Trap." *Wilmott Magazine*, Jan. 2007, pp. 83-92.

Bates, David S. "Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options." *The Review of Financial Studies*, vol. 9, no. 1, 1996, pp. 69-107.

Carr, Peter, and Dilip B. Madan. "Option Valuation Using the Fast Fourier Transform." *Journal of Computational Finance*, vol. 2, no. 4, 1999, pp. 61-73.

Cox, John C., Jonathan E. Ingersoll, and Stephen A. Ross. "A Theory of the Term Structure of Interest Rates." *Econometrica*, vol. 53, no. 2, 1985, pp. 385-407.

Feller, William. "Two Singular Diffusion Problems." *Annals of Mathematics*, vol. 54, no. 1, 1951, pp. 173-182.

Heston, Steven L. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *The Review of Financial Studies*, vol. 6, no. 2, 1993, pp. 327-343.

Hilpisch, Yves. *Derivatives Analytics with Python: Data Analysis, Models, Simulation, Calibration and Hedging*. Wiley, 2015.

Kemna, Angelien G. Z., and Antonius C. F. Vorst. "A Pricing Method for Options Based on Average Asset Values." *Journal of Banking and Finance*, vol. 14, no. 1, 1990, pp. 113-129.

Lewis, Alan L. "A Simple Option Formula for General Jump-Diffusion and Other Exponential Lévy Processes." *SSRN Electronic Journal*, 2001, papers.ssrn.com/sol3/papers.cfm?abstract_id=282110.

Merton, Robert C. "Option Pricing When Underlying Stock Returns Are Discontinuous." *Journal of Financial Economics*, vol. 3, no. 1-2, 1976, pp. 125-144.

WorldQuant University. *MScFE 622 Stochastic Modeling: Course Notes*. WorldQuant University, 2026.